

PHYS-467 Exercise: Revisiting Inference

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Exercise 1: Particle Decay

Recall the particle decay problem you saw in the lecture. We have a source that emits particles, which decay at some point. The probability of a particle decay at position x is distributed according to

$$P(x) = \frac{1}{\lambda} \exp\left(-\frac{x}{\lambda}\right). \quad (1)$$

Our goal is to infer the λ^{true} , the “ground truth” value of λ , given some data points x that were obtained from the above distribution. The dataset contains the exact x_i (suppose we have an infinite detector), i.e. we use data points $\underline{x} = \{x_i\}_{i=1, \dots, N}$ with

$$x_i \sim P(x_i | \lambda^{\text{true}}) = \frac{1}{\lambda^{\text{true}}} e^{-x_i / \lambda^{\text{true}}} \text{ i.i.d.} \quad (2)$$

In this exercise, we want to construct the maximum likelihood and MAP estimators for finding λ^{true} .

1. Write the likelihood for a complete dataset $\underline{x}$, i.e. $\mathcal{L}(\lambda) = P(\underline{x} | \lambda)$.

Solution:

$$\mathcal{L}(\lambda) = P(\underline{x} | \lambda) = \prod_{i=1}^N P(x_i | \lambda) = \frac{1}{\lambda^N} e^{-\frac{1}{\lambda} \sum_{i=1}^N x_i}$$

2. Write the posterior distribution $P(\lambda | \underline{x})$ in terms of the prior and the likelihood.

Solution:

$$P(\lambda | \underline{x}) = \frac{P(\underline{x} | \lambda) P(\lambda)}{P(\underline{x})} = \frac{P(\underline{x} | \lambda) P(\lambda)}{\int d\lambda P(\underline{x} | \lambda) P(\lambda)}$$

3. Compute the maximum likelihood estimator $\hat{\lambda}_{\text{ML}} = \arg \max_{\lambda} \log P(\underline{x} | \lambda)$.

Solution: To obtain the ML-estimator we maximize the log-likelihood l of λ (or equivalently, minimize the negative log-likelihood).

$$l(\lambda) = \log P(\underline{x} | \lambda) = -N \log(\lambda) - \frac{1}{\lambda} \sum_{i=1}^N x_i$$

To find the maximum we set the derivative to zero and then solve for λ

$$\begin{aligned} l'(\lambda) &= -\frac{N}{\lambda} + \frac{1}{\lambda^2 \sum_i x_i} = 0, \\ \Rightarrow \lambda &= \frac{1}{N} \sum_{i=1}^N x_i. \end{aligned}$$

(and we also check that this is a maximum by computing the second derivative).

4. Let's assume we have some prior information about the distribution of the λ , e.g. the exponential prior $P(\lambda) = -\frac{1}{\lambda_0} e^{-\frac{\lambda}{\lambda_0}}$, where λ_0 is the mean value of this distribution. Compute the maximum aposterior estimator $\hat{\lambda}_{\text{MAP}} = \arg \max_{\lambda} \log[P(\underline{x}|\lambda)P(\lambda)]$ in terms of the data and the prior λ_0 . What happens to the expression when we increase the number of data samples as $N \rightarrow \infty$?

Solution: To obtain the MAP-estimator we maximize the log of the likelihood and the prior l of λ .

$$l(\lambda) = \log[P(\underline{x}|\lambda)P(\lambda)] = -N \log(\lambda) - \frac{1}{\lambda} \sum_{i=1}^N x_i - \frac{\lambda}{\lambda_0} - \log \lambda_0$$

To find the maximum we set the derivative to zero, ignoring all constant terms in λ and then solve for λ

$$\begin{aligned} l'(\lambda) &= -\frac{N}{\lambda} + \frac{1}{\lambda^2 \sum_i x_i} - \frac{1}{\lambda_0} = 0, \\ \Rightarrow \lambda &= \frac{N\lambda_0}{2} \left(-1 \pm \sqrt{1 + \frac{4}{N^2\lambda_0} \sum_{i=1}^N x_i} \right) \end{aligned}$$

Since we want $\lambda > 0$, we select the $+$ sign. If we take the $N \rightarrow \infty$ limit where we have more and more data available we can observe that $\sqrt{1+a} = 1 + \frac{1}{2}a + O(a^2)$. By this simplification, we get

$$\lambda = \frac{N\lambda_0}{2} \left(\frac{2}{N^2\lambda_0} \sum_{i=1}^N x_i \right) = \frac{1}{N} \sum_{i=1}^N x_i$$

which shows us that if we have an infinite amount of data available, we do not take the prior into consideration anymore.

Exercise 2: Bent Coin

We now want to look at (un)fair coin tosses. Imagine that we have a coin and the probability that the coin shows the head is p , and $1-p$ for tail, a Bernoulli distribution. Let $\underline{s} = (s_1, s_2, \dots, s_N)$ be a sequence of tosses that you observed from the same coin and let $s_i = 1$ mean that a head was tossed in the i^{th} trial.

1. We now want to determine if the coin was fair, i.e. whether $p = 1/2$ or not, for which we need to find an estimator for p . For this, compute the likelihood $P(\underline{s}|p)$ and derive the maximum likelihood estimator $\hat{p}_{\text{ML}}$.

Solution: The likelihood is

$$P(\underline{s}|p) = \prod_{i=1}^N p^{s_i} (1-p)^{(1-s_i)} = p^{\sum_{i=1}^N s_i} (1-p)^{N - \sum_{i=1}^N s_i}$$

which leads to the likelihood estimator

$$\begin{aligned} l(p) &= \log P(\underline{s}|p) = \sum_{i=1}^N s_i \log p - (N - \sum_{i=1}^N s_i) \log(1-p) \\ l'(p) &= \frac{1}{p} \sum_{i=1}^N s_i - \frac{N}{1-p} + \frac{1}{1-p} \sum_{i=1}^N s_i = 0 \\ \Rightarrow p &= \frac{1}{N} \sum_{i=1}^N s_i \end{aligned}$$

2. You take part in a game where you observe the first 7 tosses of a possibly bent coin, as $\underline{s} = (1, 1, 0, 0, 1, 1, 1)$. As part of the game, you have to decide whether you want to bet on the next two tosses being head. If you are correct you win the game. To make an informed decision on your bet, evaluate $P(s_8 = 1, s_9 = 1 | \hat{p}_{\text{ML}})$ and determine what you should bet on.

Solution: We have $\hat{p}_{\text{ML}} = \frac{1}{7} \sum_{i=1}^7 s_i = 5/7$ from before. Therefore, the probability that we see two heads following the given sequence is $(5/7)^2 = 0.510\dots$. We should therefore bet on two heads!

3. Assume that you obtained knowledge on the game that the coins are usually not bent uniformly in the range of $[0, 1]$, but are far more likely to be bent close to a fair coin. You model this prior as $P(p) = 6p(1 - p)$ and additionally add that $p \in [0, 1]$. Take a Bayesian approach to your bet and compute

$$\int P(s_8 = 1, s_9 = 1 | p) P(p | \underline{s}) dp$$

to determine what you want to bet on.

Solution: We first compute the posterior

$$P(p | \underline{s}) = \frac{P(\underline{s} | p) P(p) \theta(p \in [0, 1])}{\int_0^1 P(\underline{s} | p) P(p) dp}$$

where θ is the indicator function to determine that p is in the correct range. For our concrete setting this gives

$$P(p | \underline{s}) = \frac{p^5(1 - p)^2 6p(1 - p) \theta(p \in [0, 1])}{\int_0^1 p^5(1 - p)^2 6p(1 - p) dp} = \frac{p^6(1 - p)^3 6 \theta(p \in [0, 1])}{1/140}$$

Inserting this into the integral

$$\int P(s_8 = 1, s_9 = 1 | p) P(p | \underline{s}) dp = \int p^2 P(p | \underline{s}) dp = \int_0^1 p^8(1 - p)^3 6 \cdot 140 dp = 14/33 = 0.4242\dots$$

With this special prior knowledge, we should therefore not bet on two heads occurring!